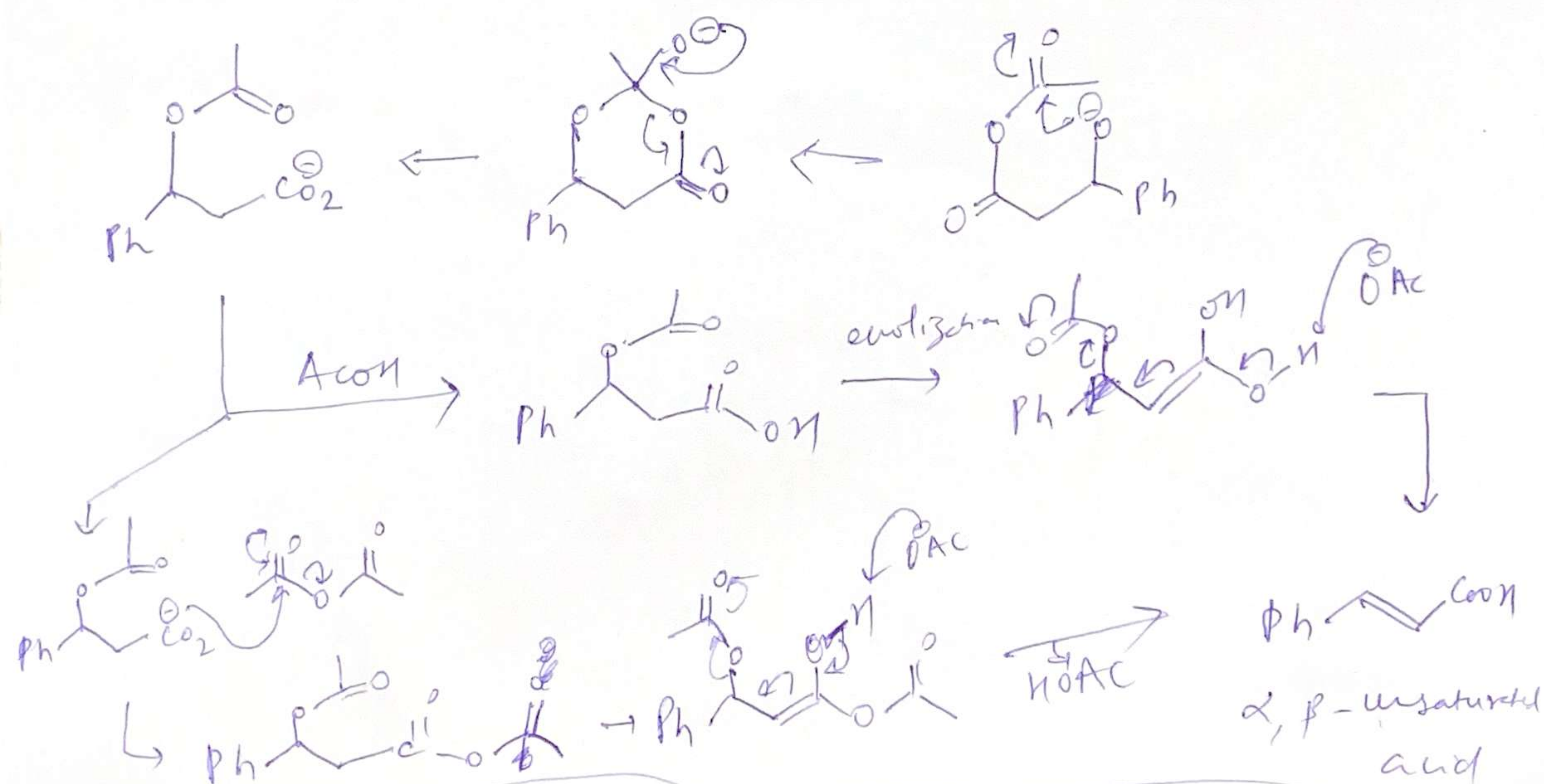
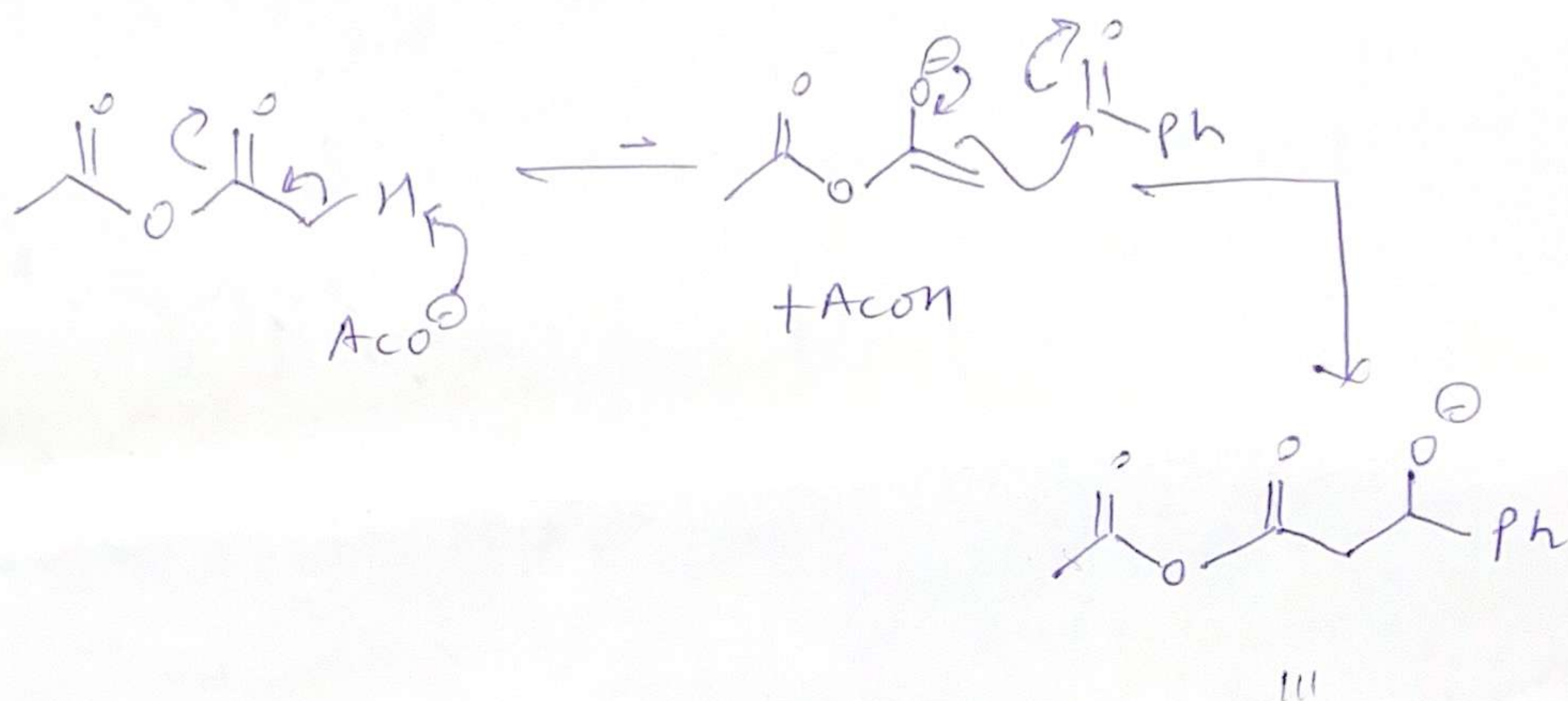
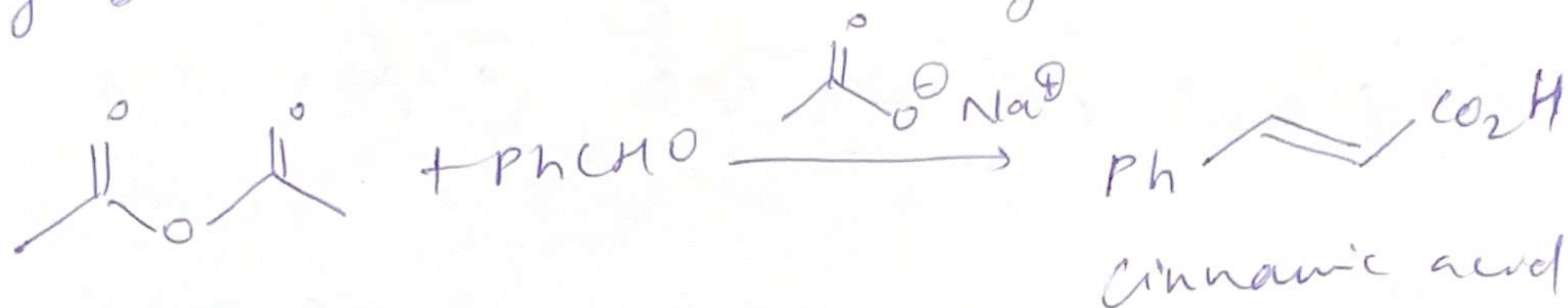
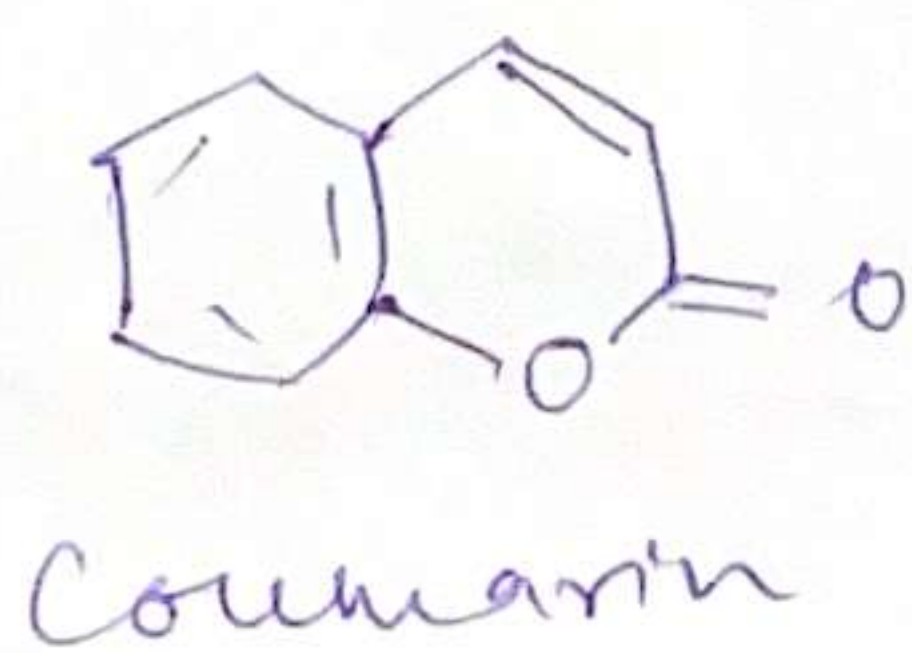
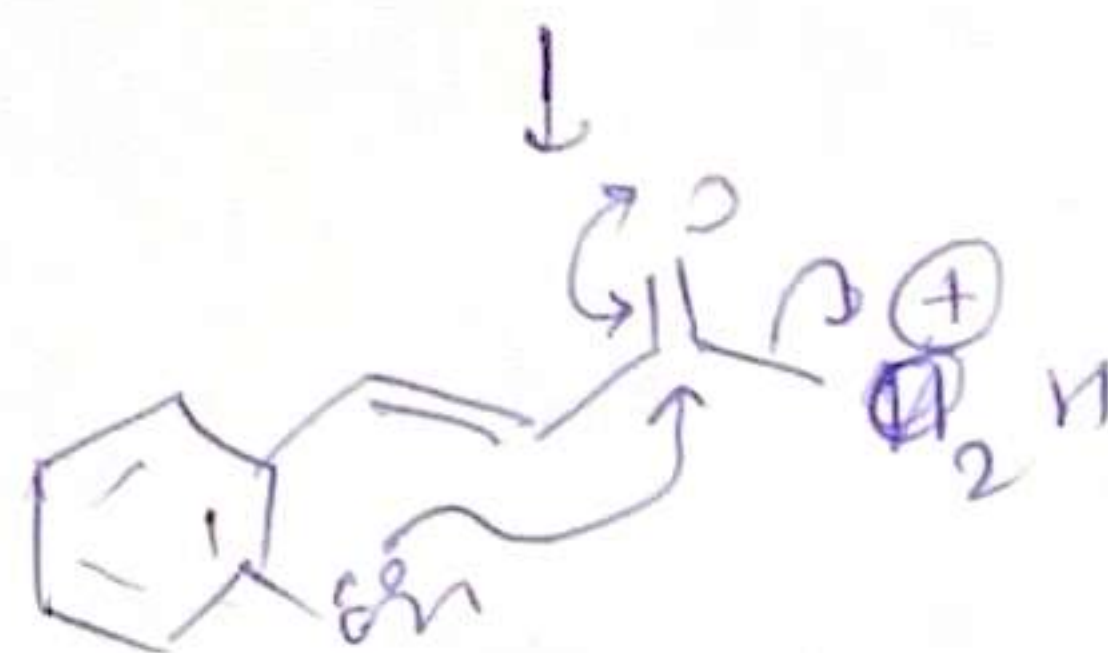
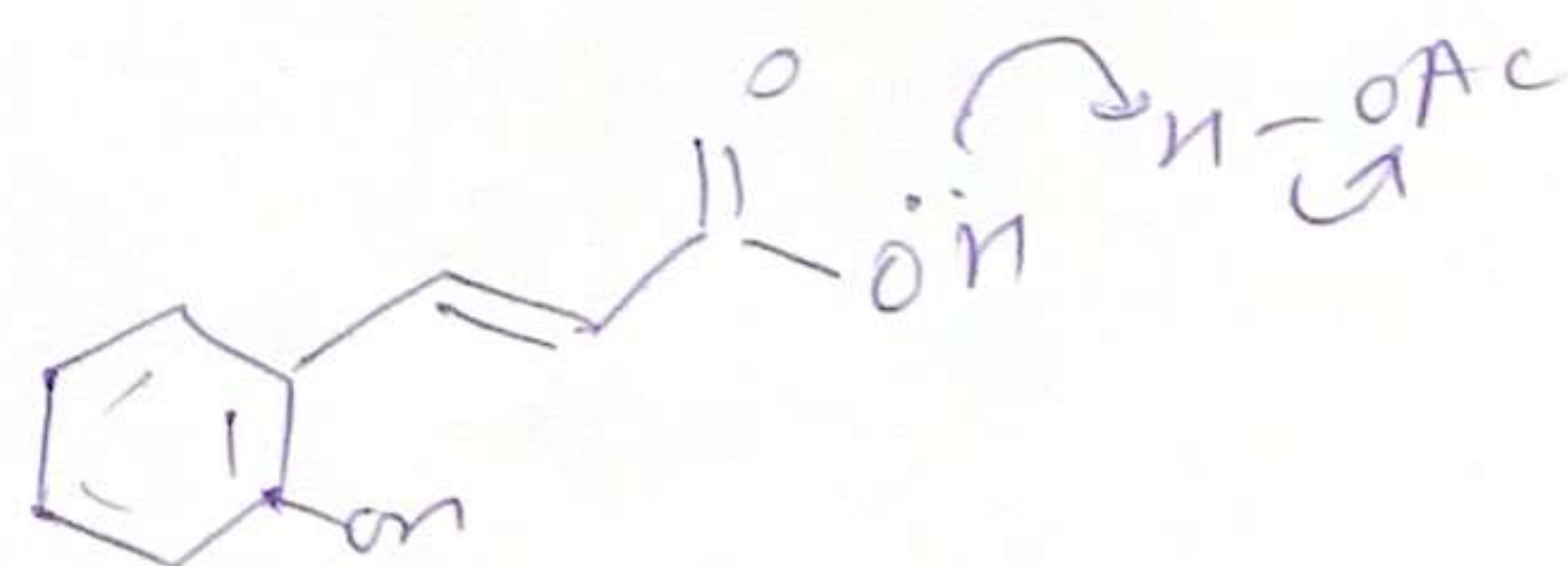
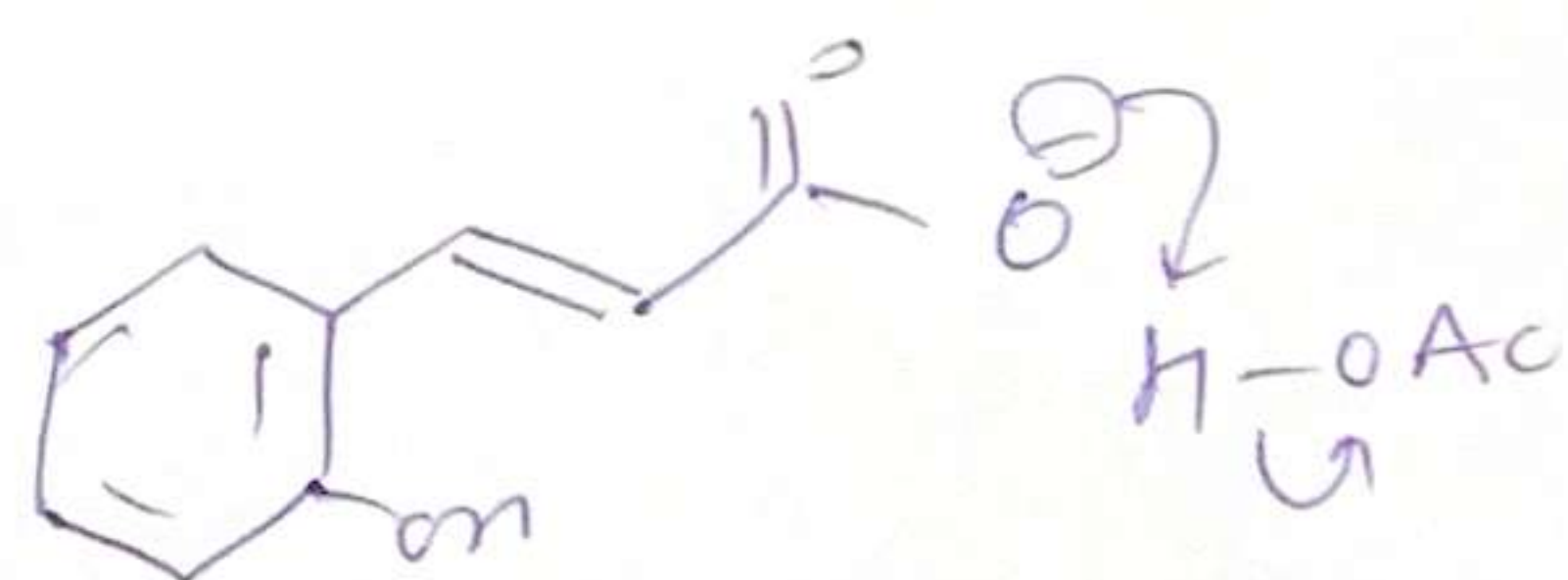
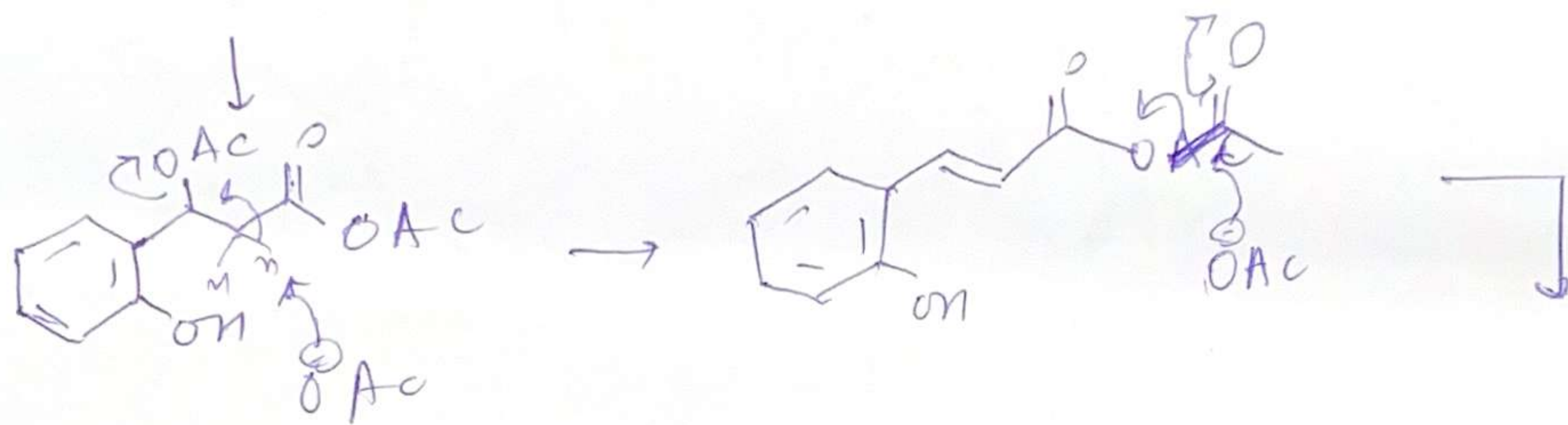
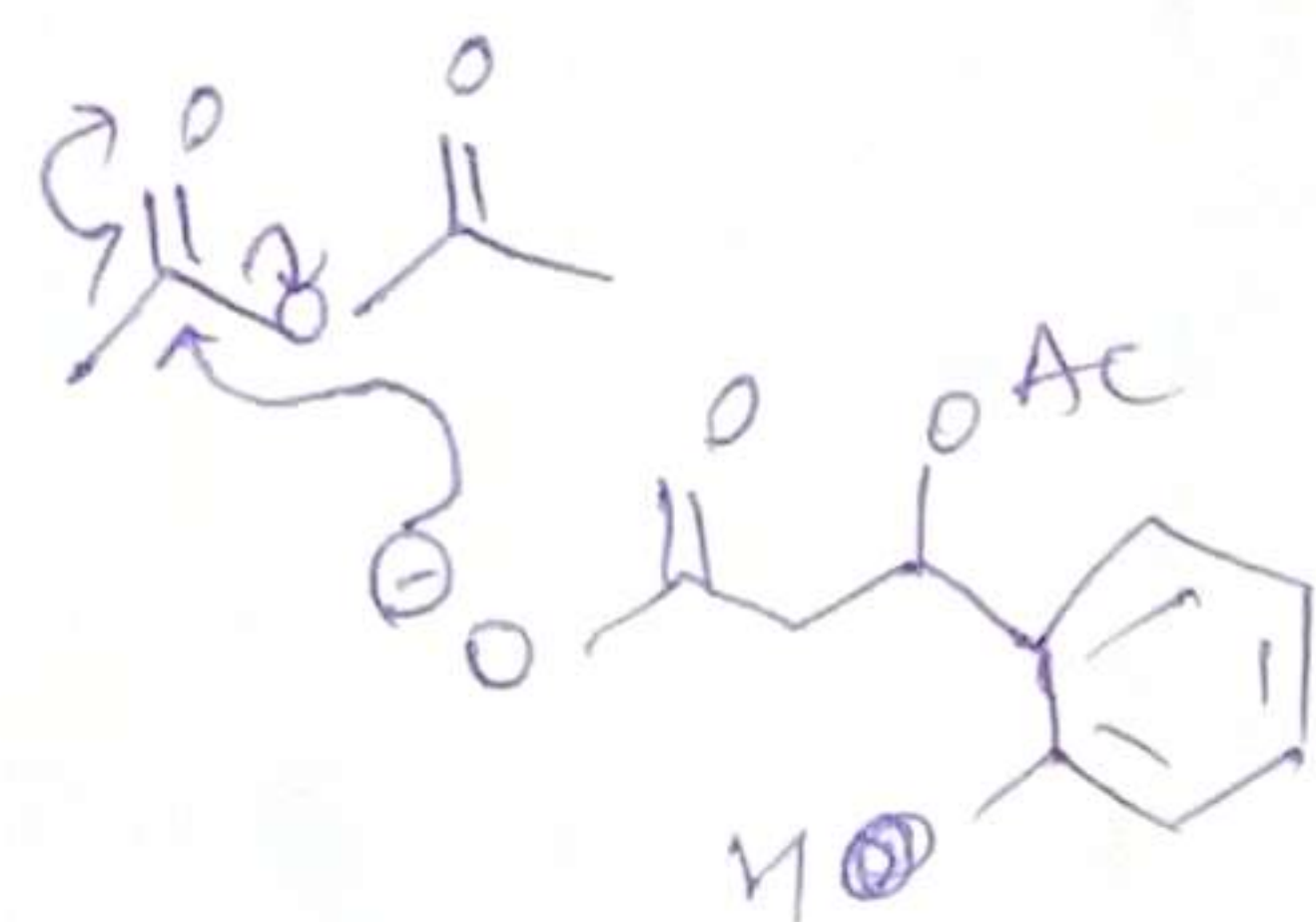
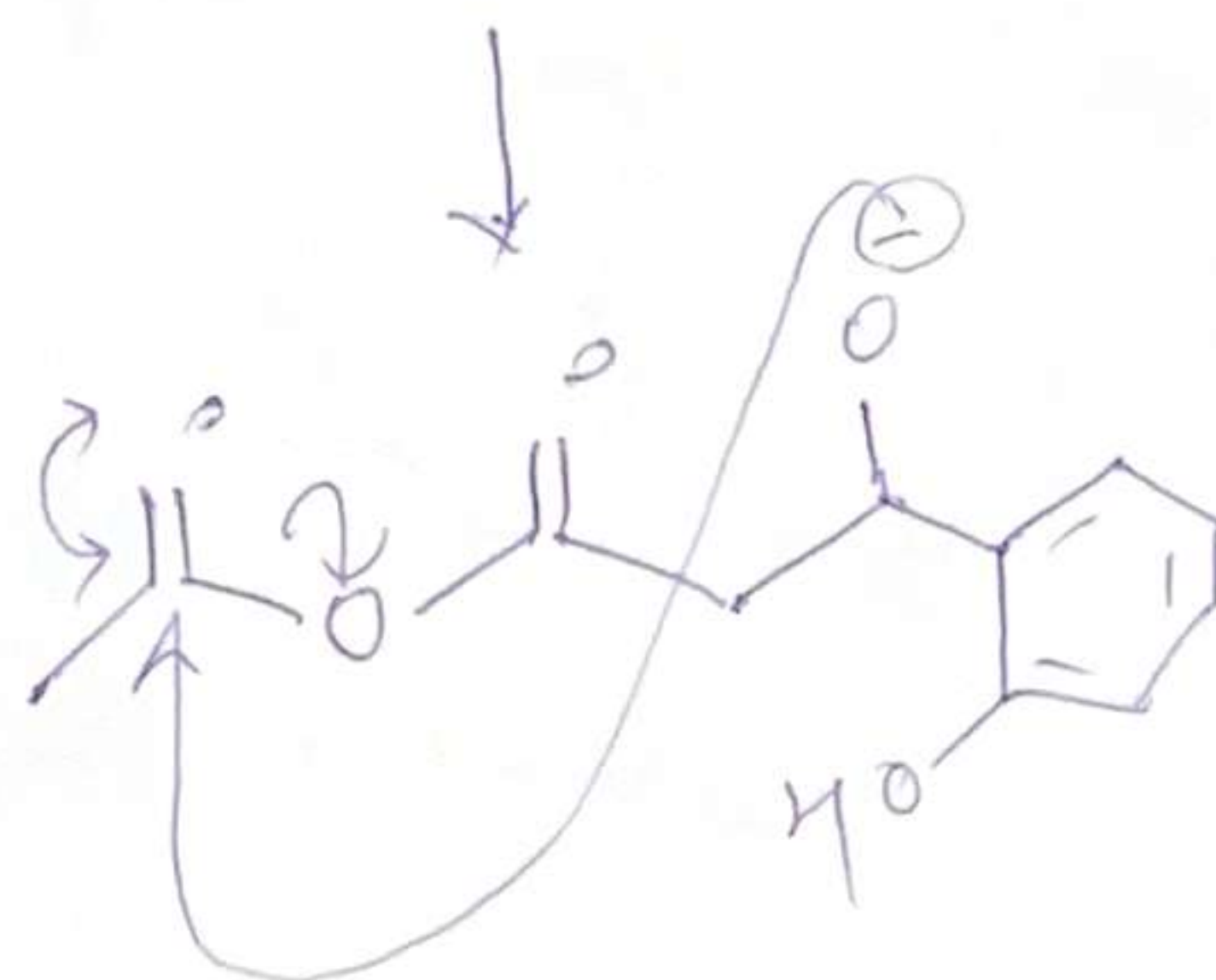
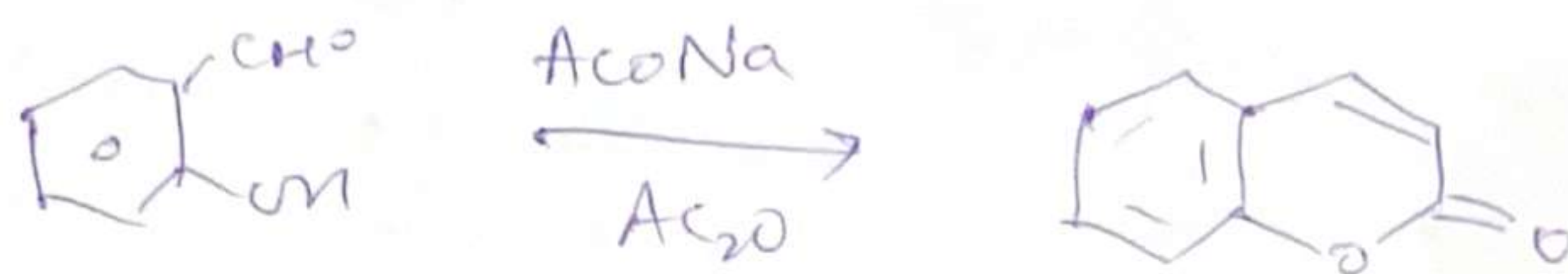


Perkin Reaction

Acid anhydride + Non-enolizable aldehyde + weak base usually the salt of acid (chosen in such a way so that Nu^- attack on anhydride does no harm)



Synthesis of coumarin from salicylaldehyde

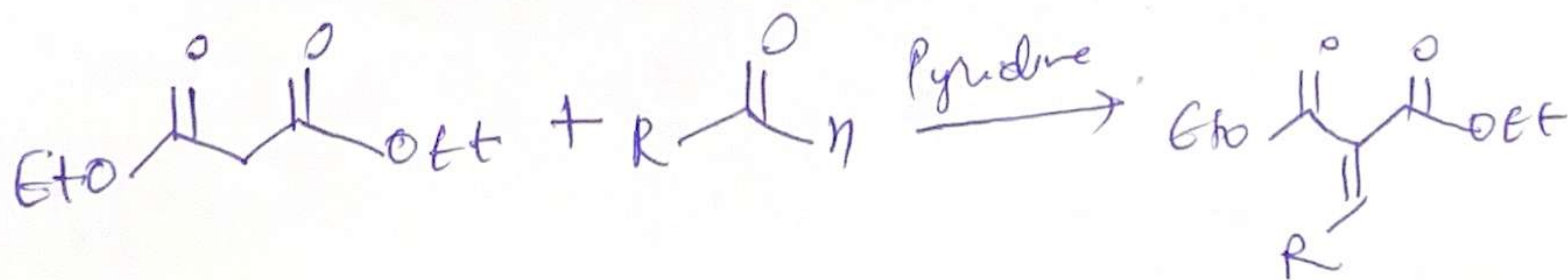
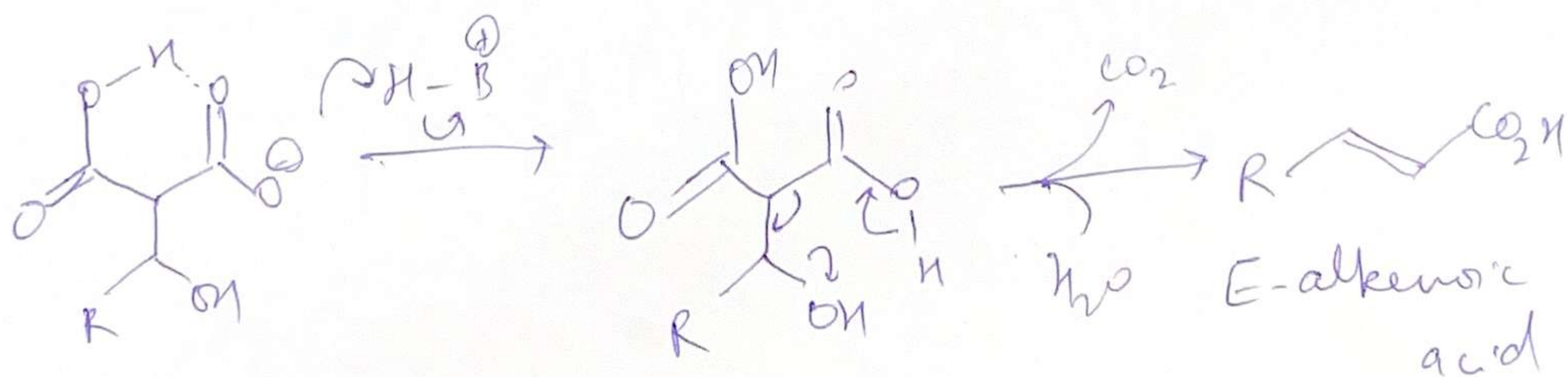
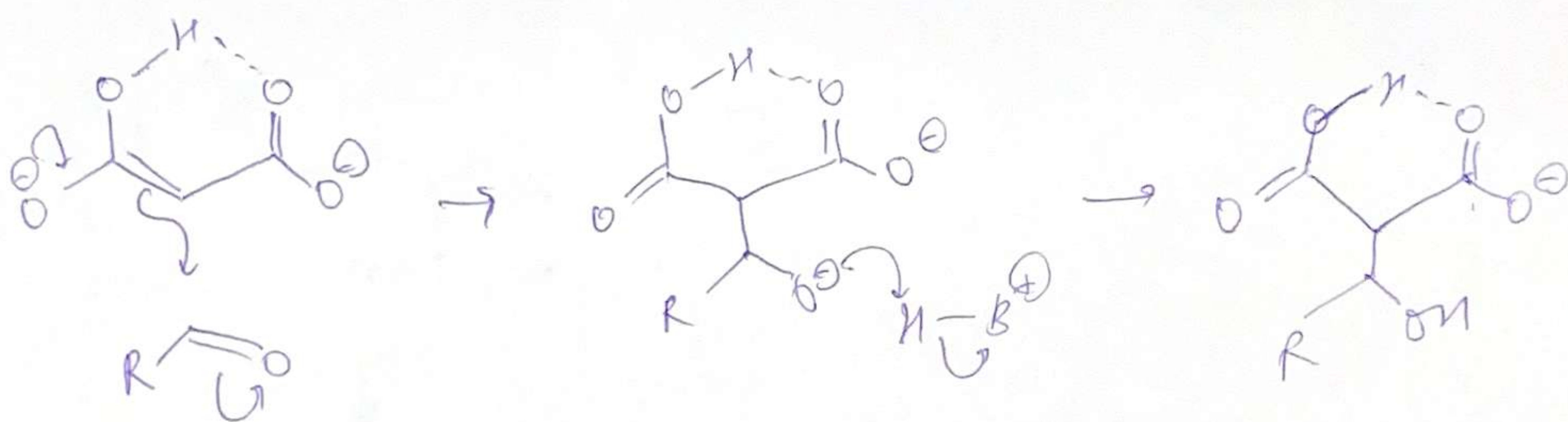
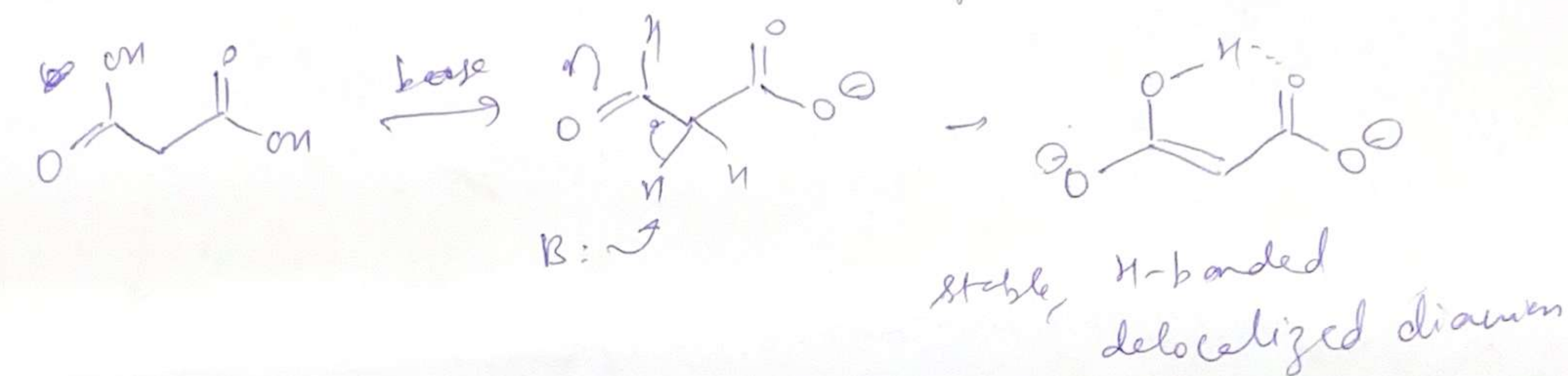
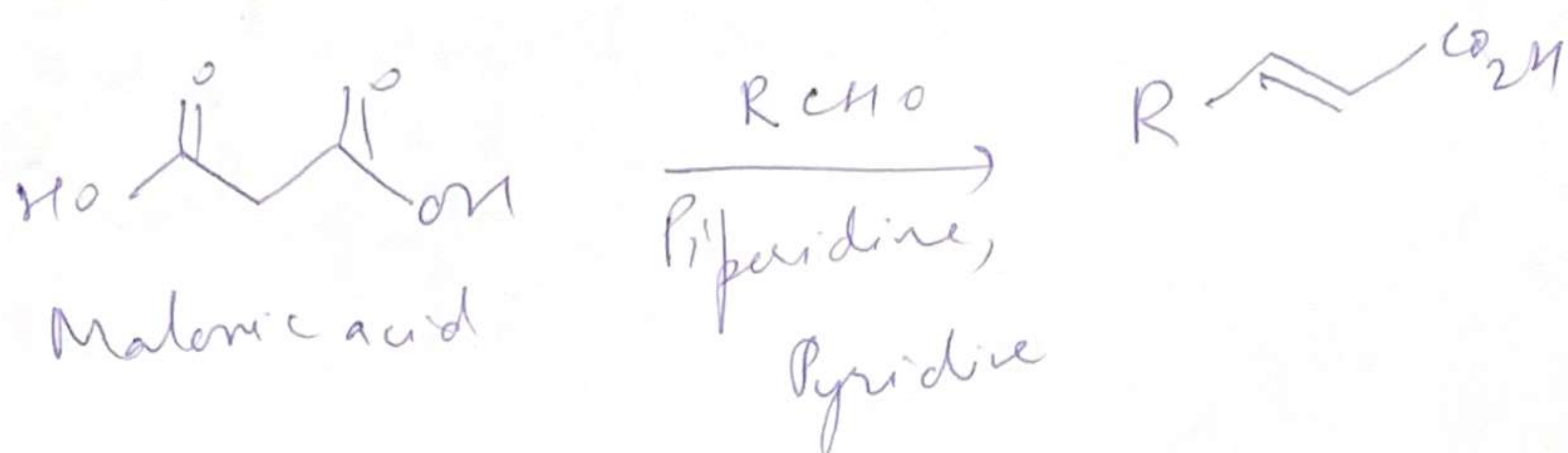


Knoevenagel Condensation

catalyst + weakly basic

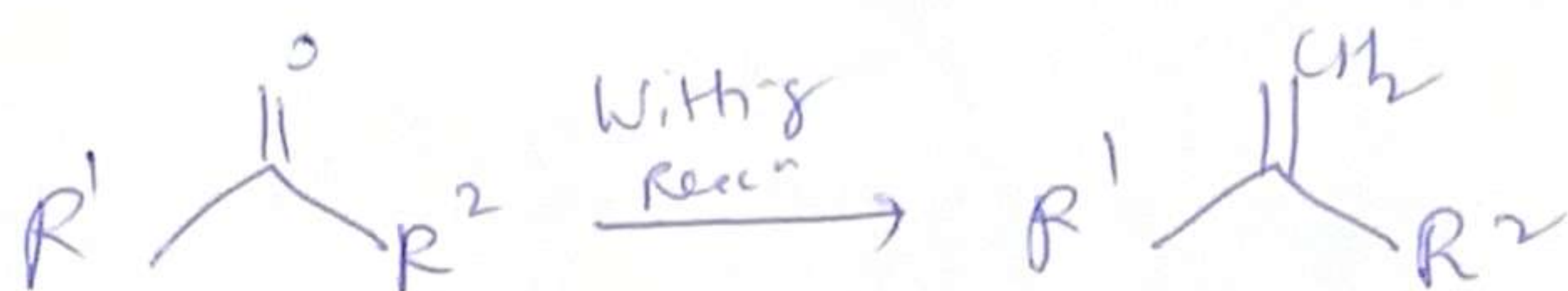
Active-H: Nitromethane, ethyl acetoacetate, diethyl malonate, cyanoacetylene

Nu⁺ addⁿ of an active hydrogen compound to a carbonyl group followed by dehydration reacⁿ.
Product is often α, β -unsaturated ketone.

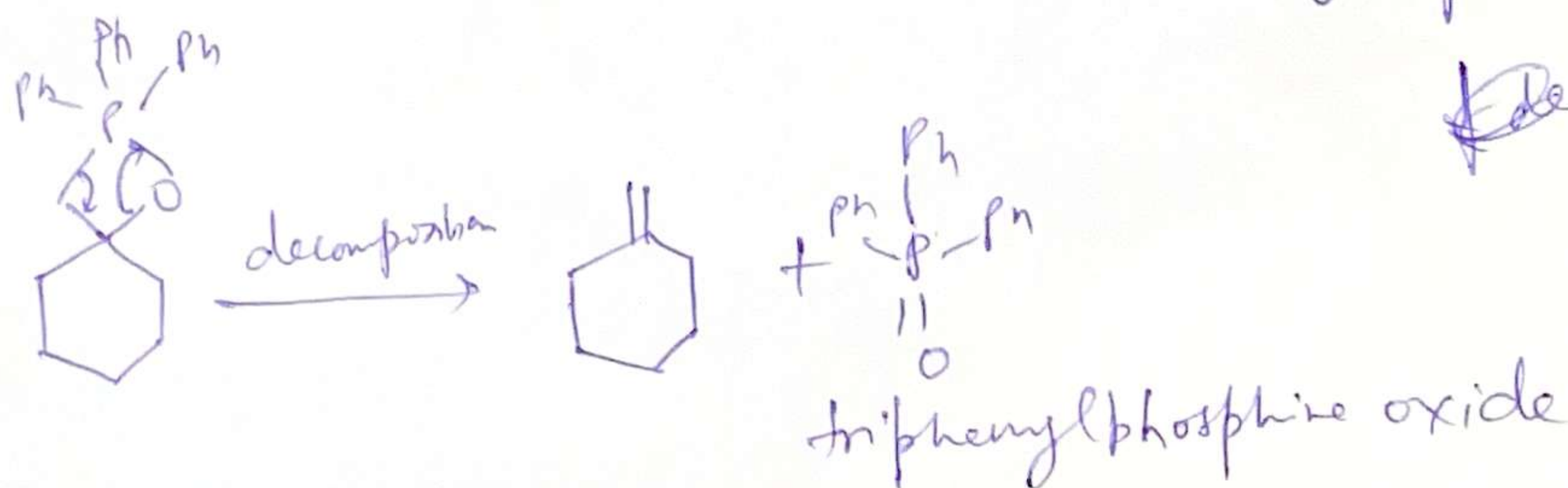
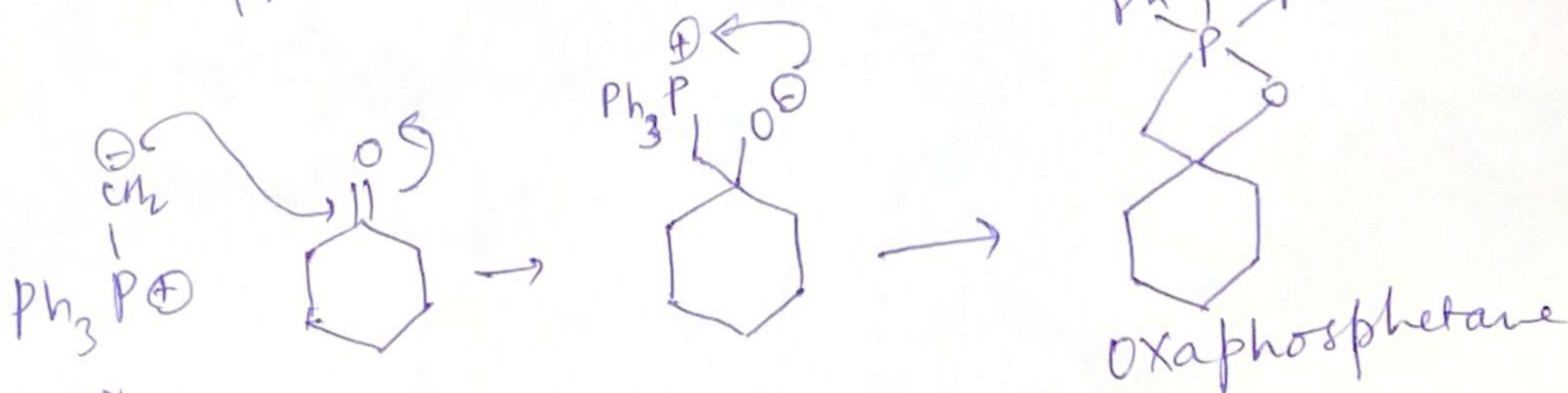
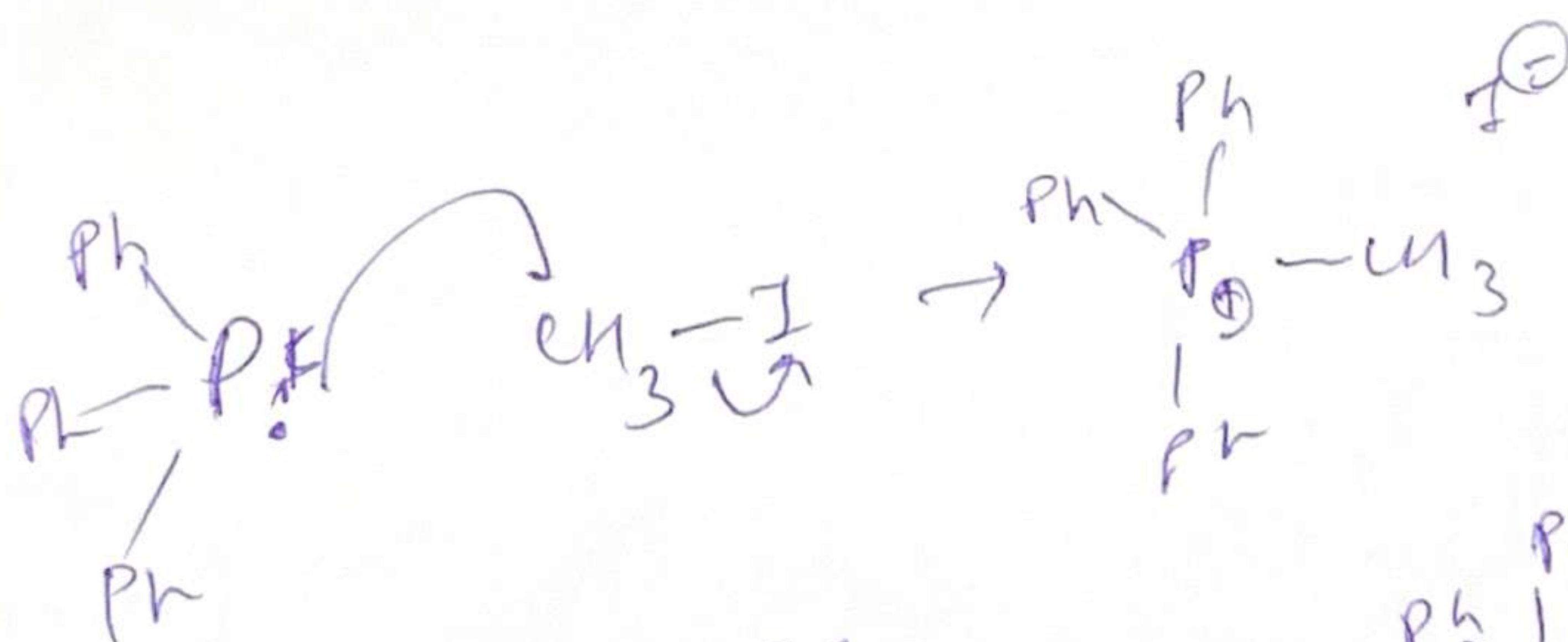
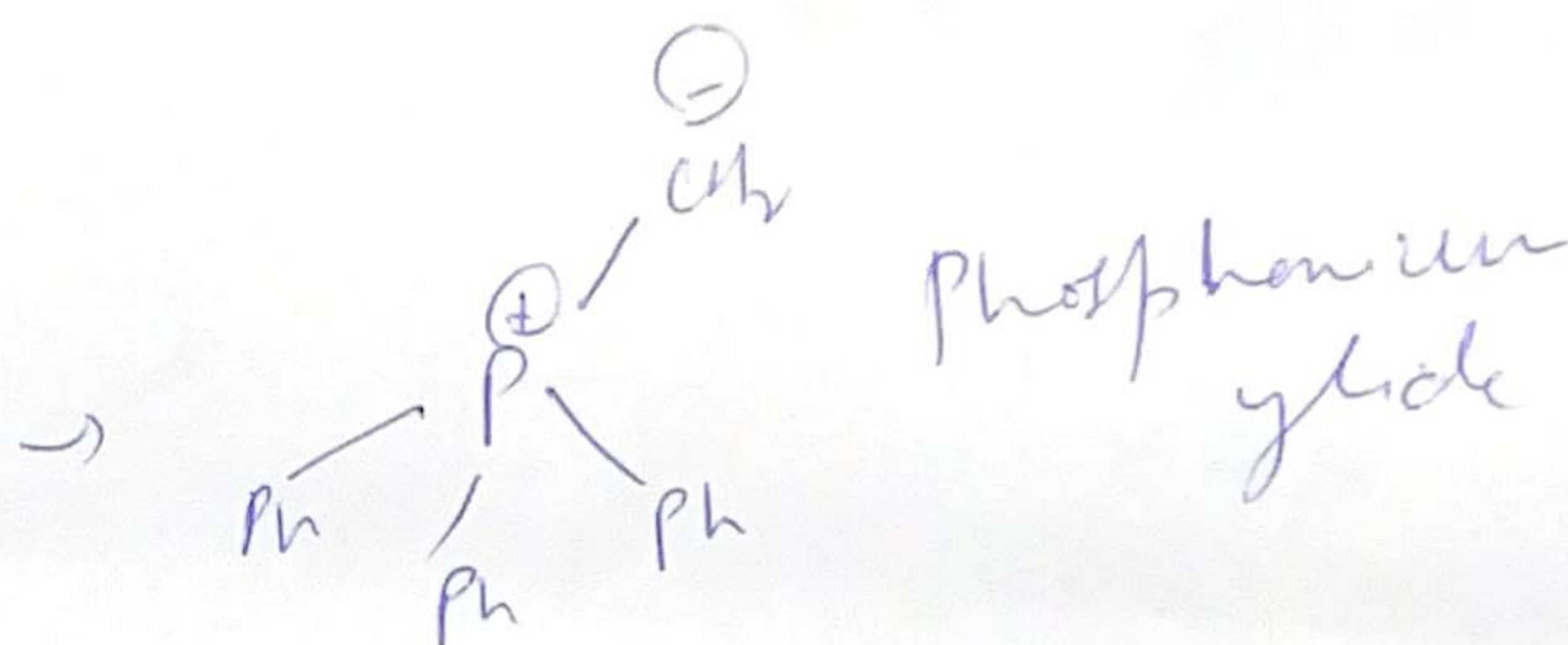
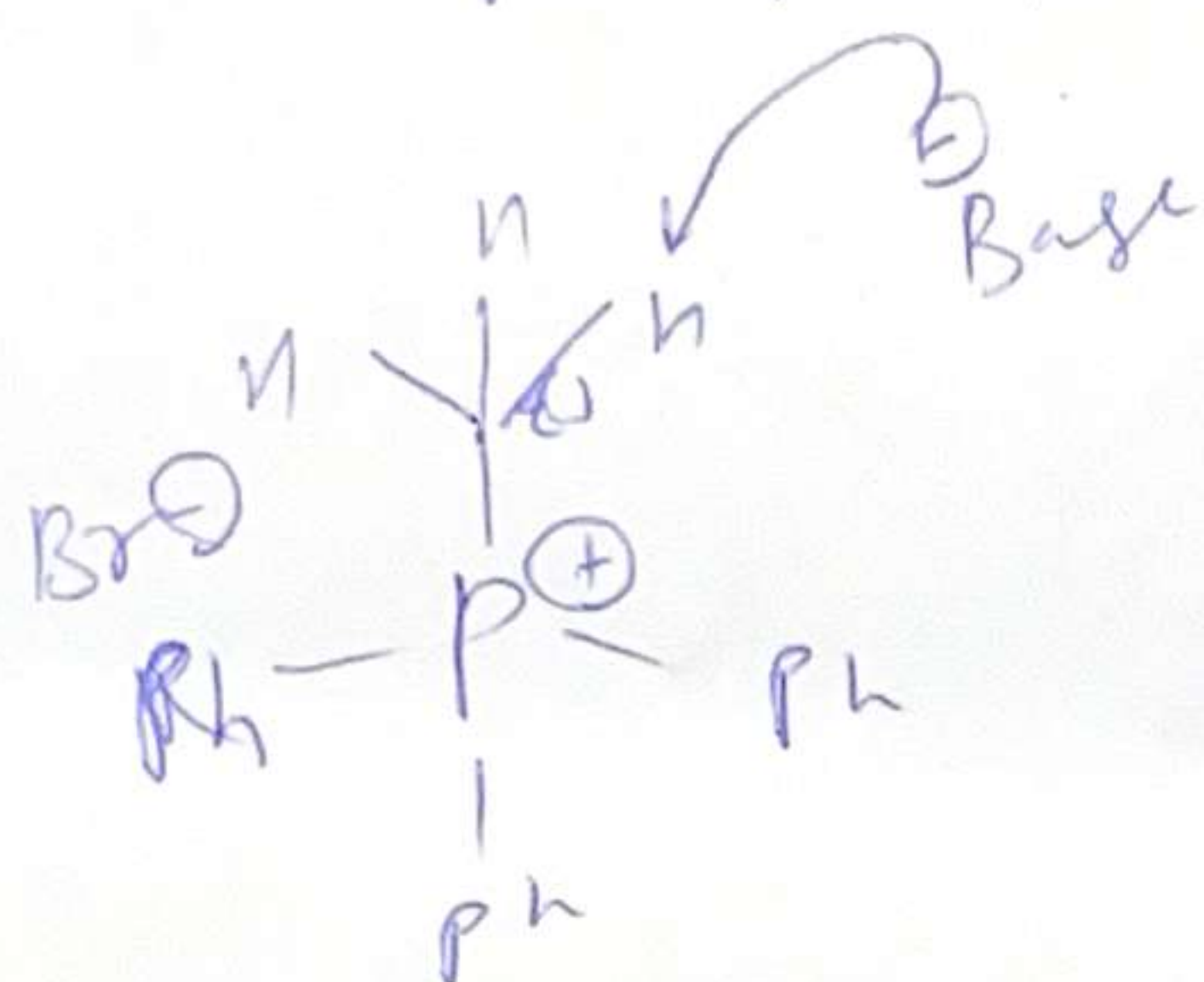
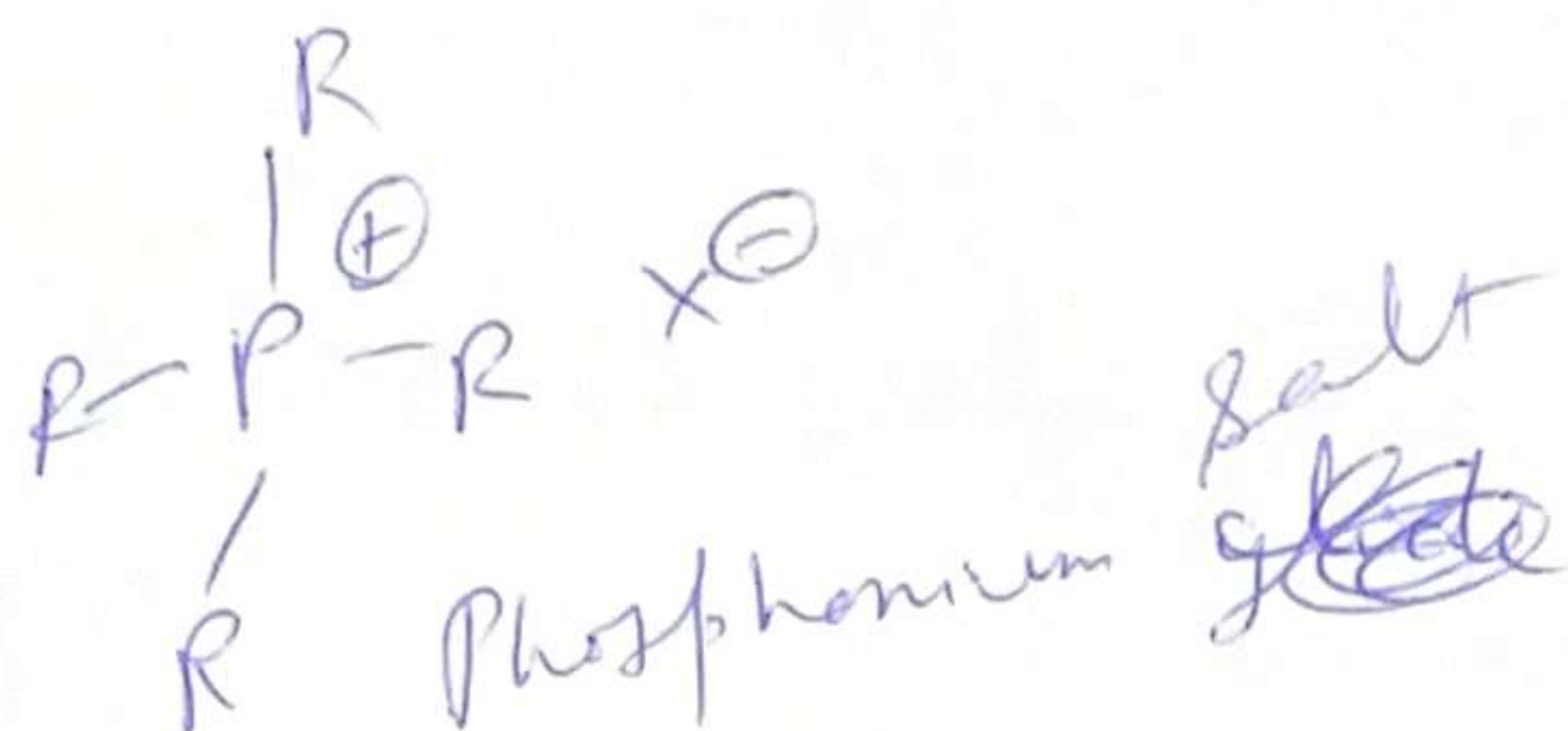


Wittig Reaction

10



Substitution of a $C=C$ bond for a $C=O$ bond
 Reaction betw. Carbonyl compd. (aldehyde/ketone only) & a phosphonium ylide.



P has strong affinity for O - ym

Gingnard Reacⁿ

Gingnard Reagent

organomagnesium halide



Aldehydes / ketones

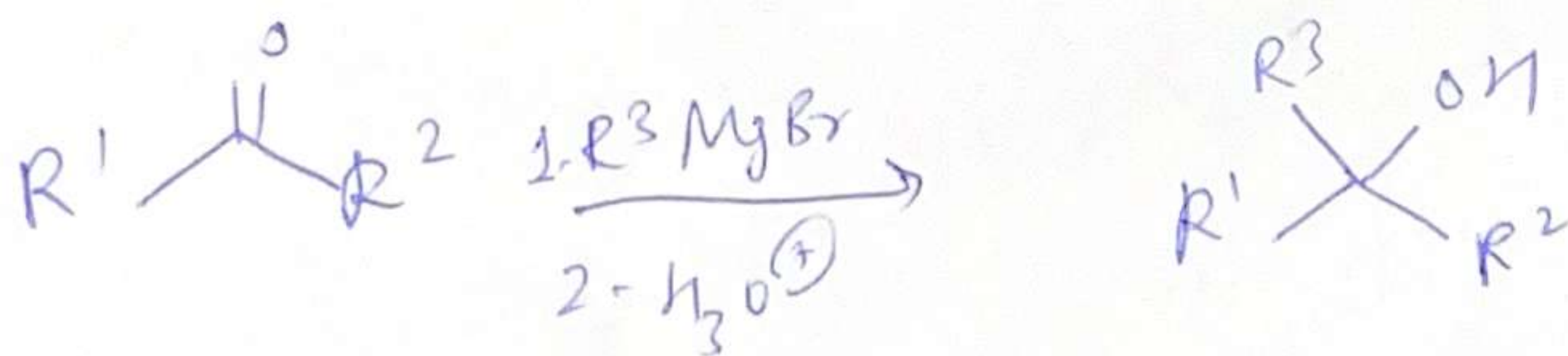
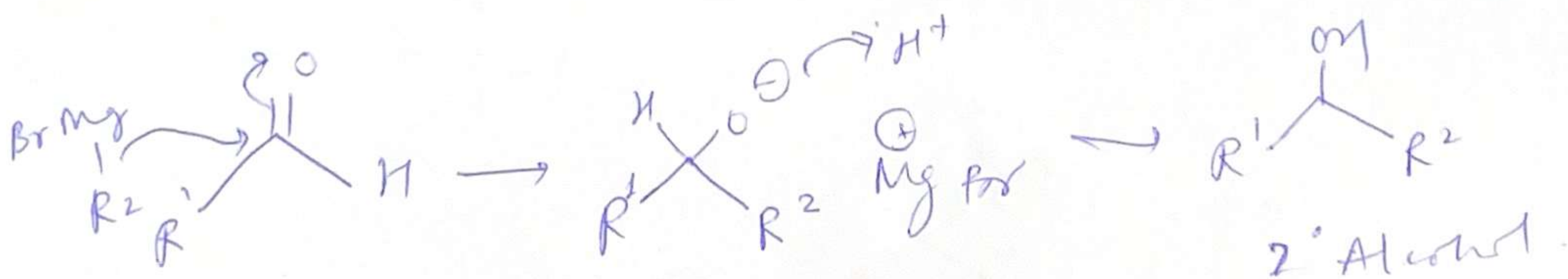
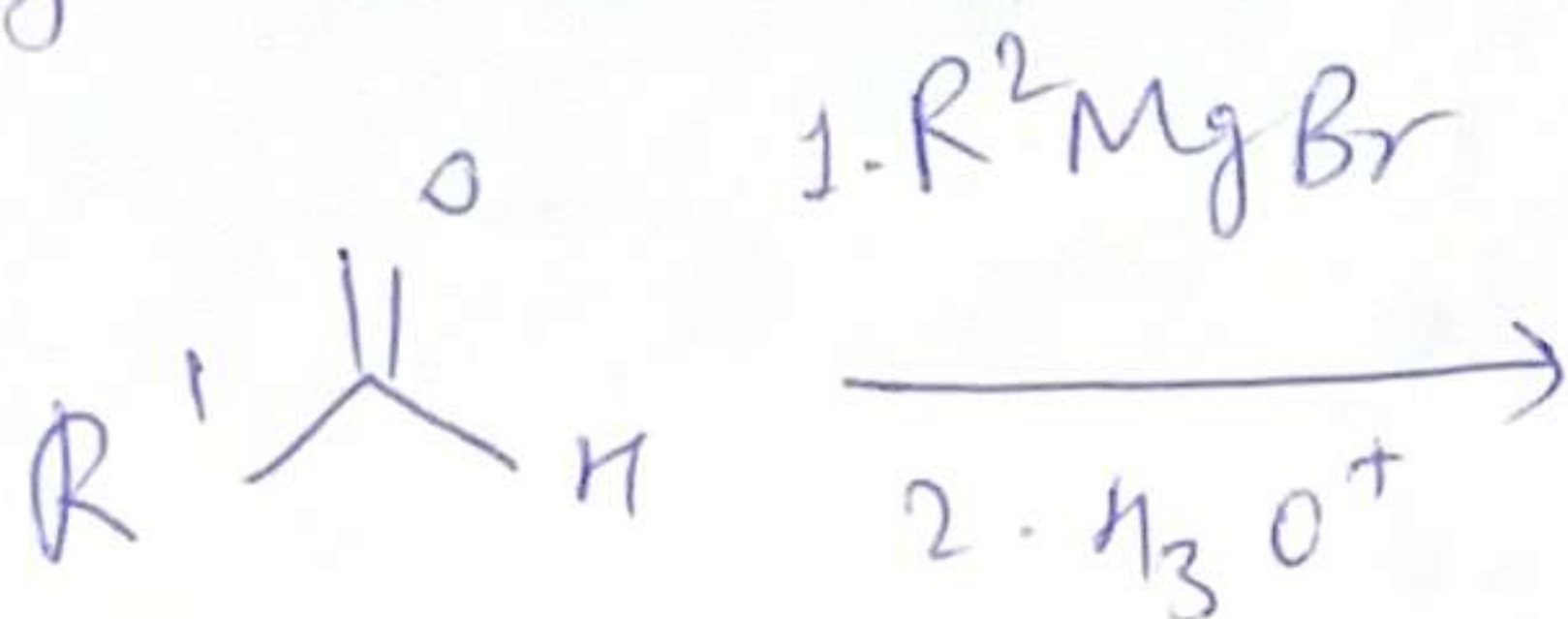
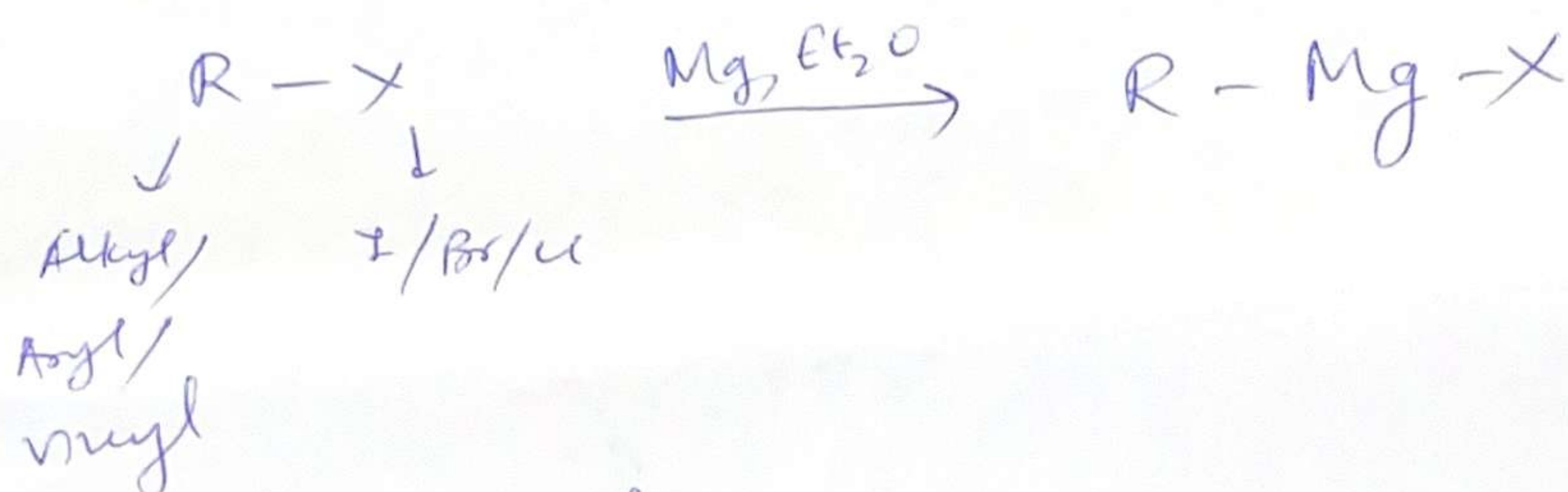


2° Alcohol



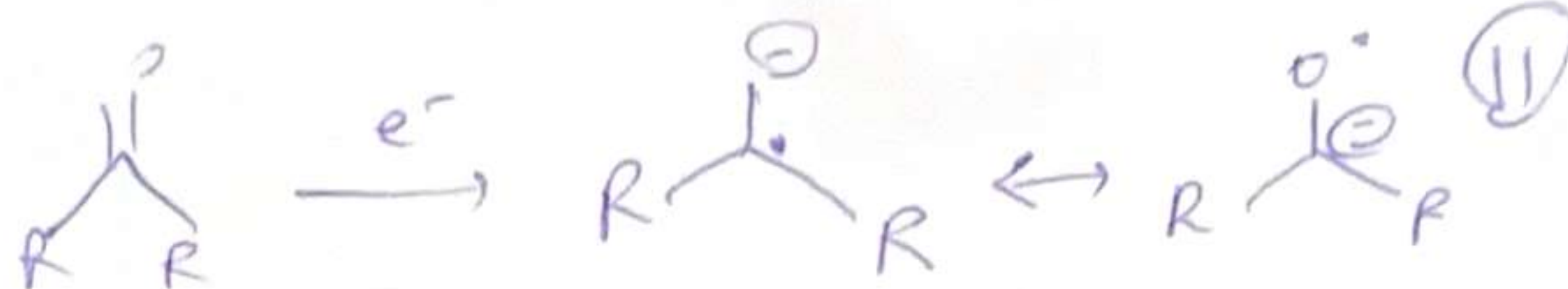
3° Alcohol

Can also work with esters, nitriles, epoxides leading to different products.

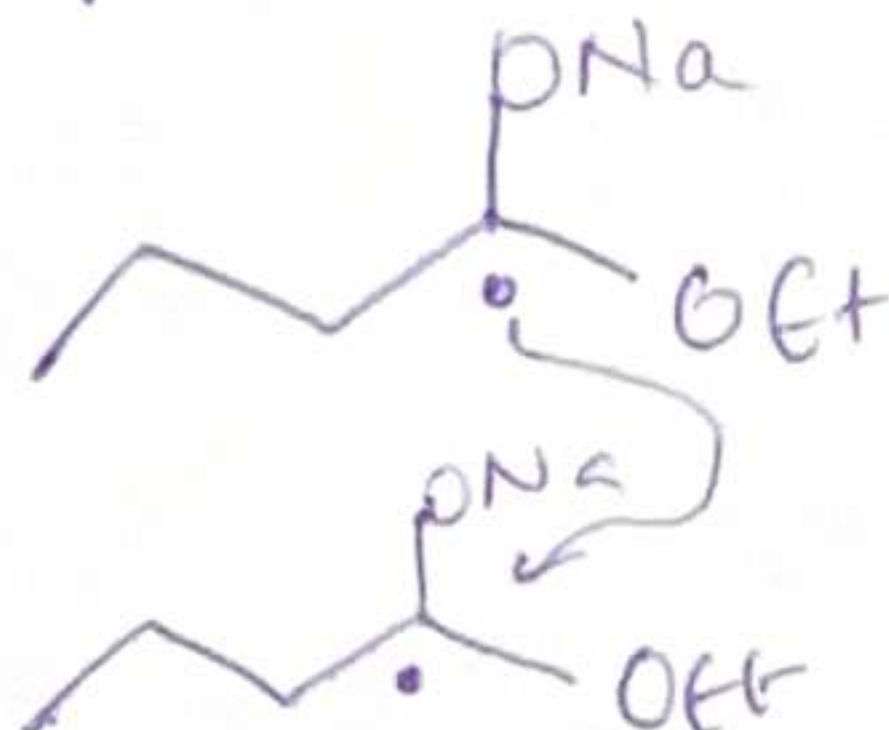
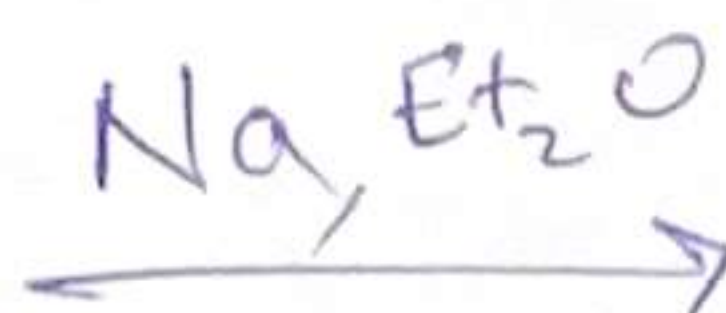
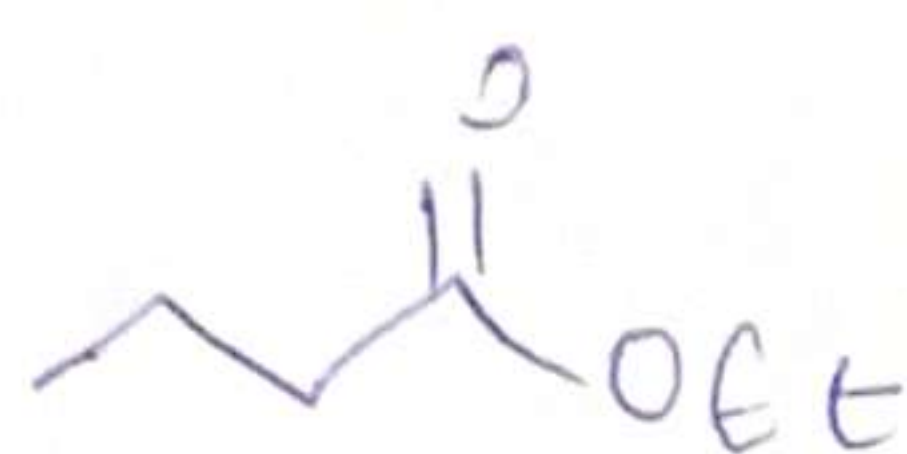


Acyloin Condensation

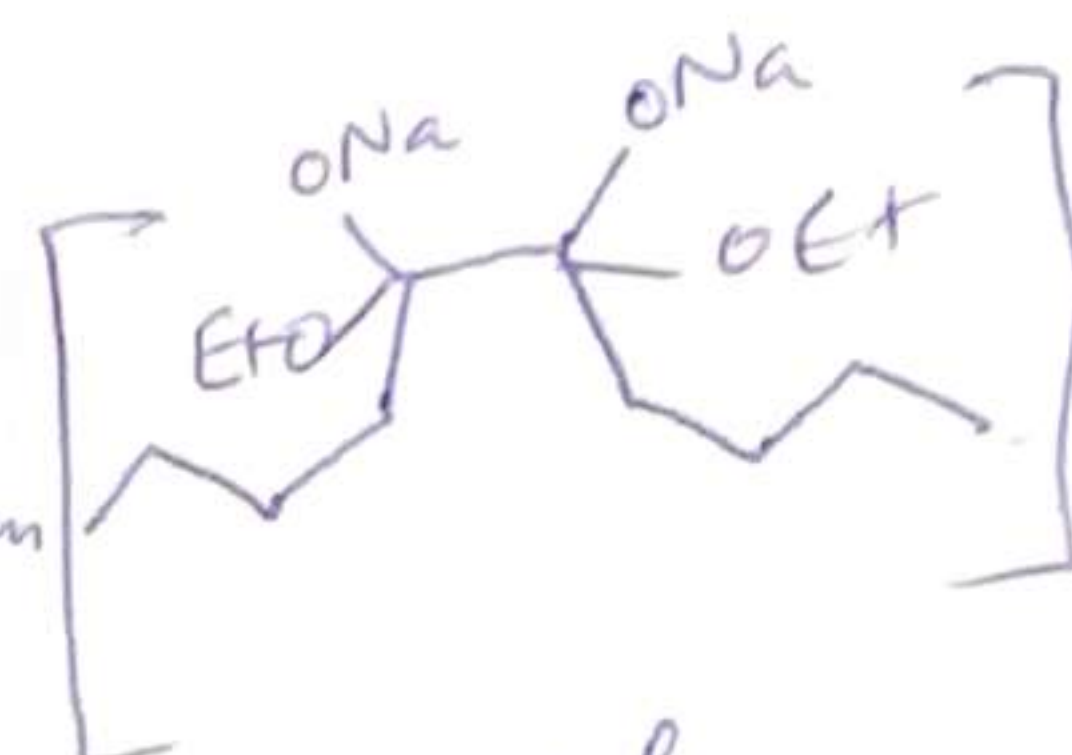
→ α -hydroxy ketone



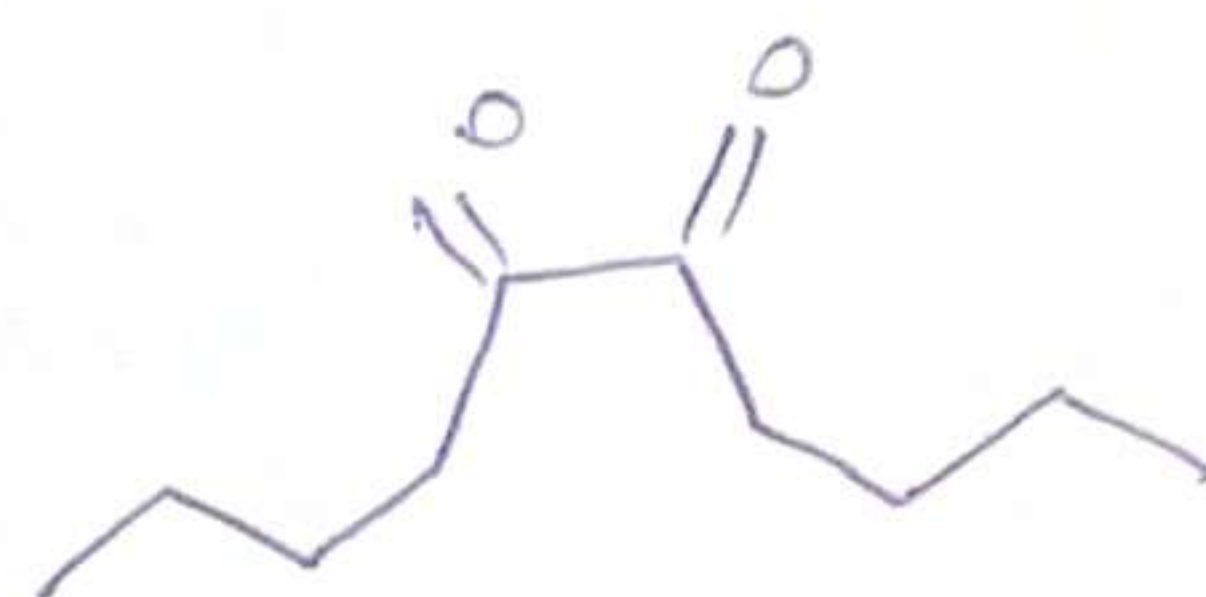
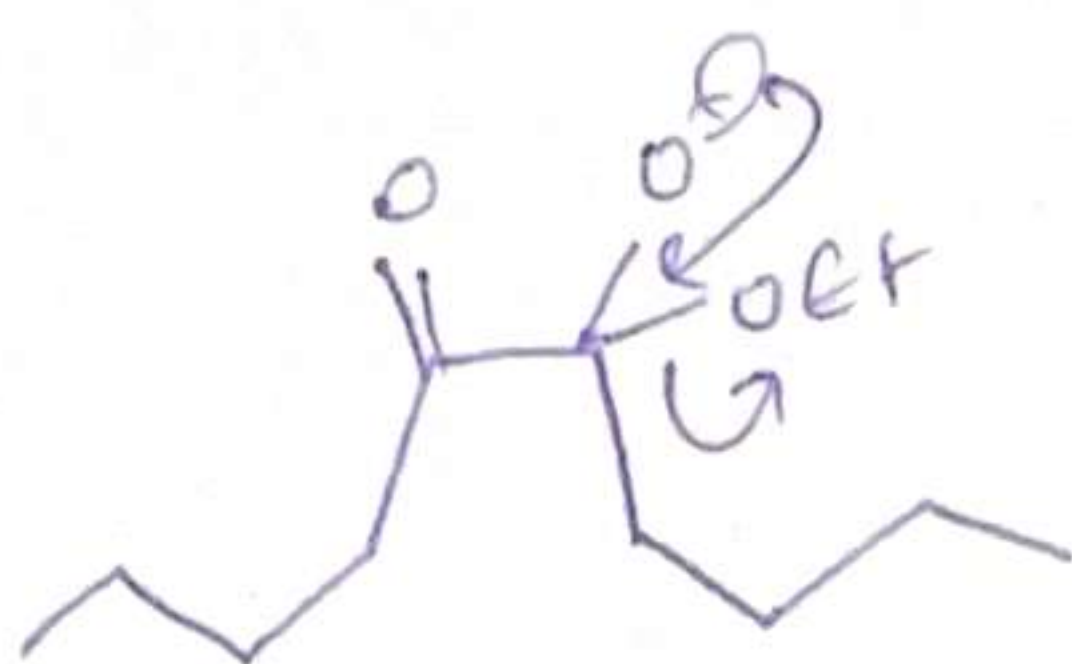
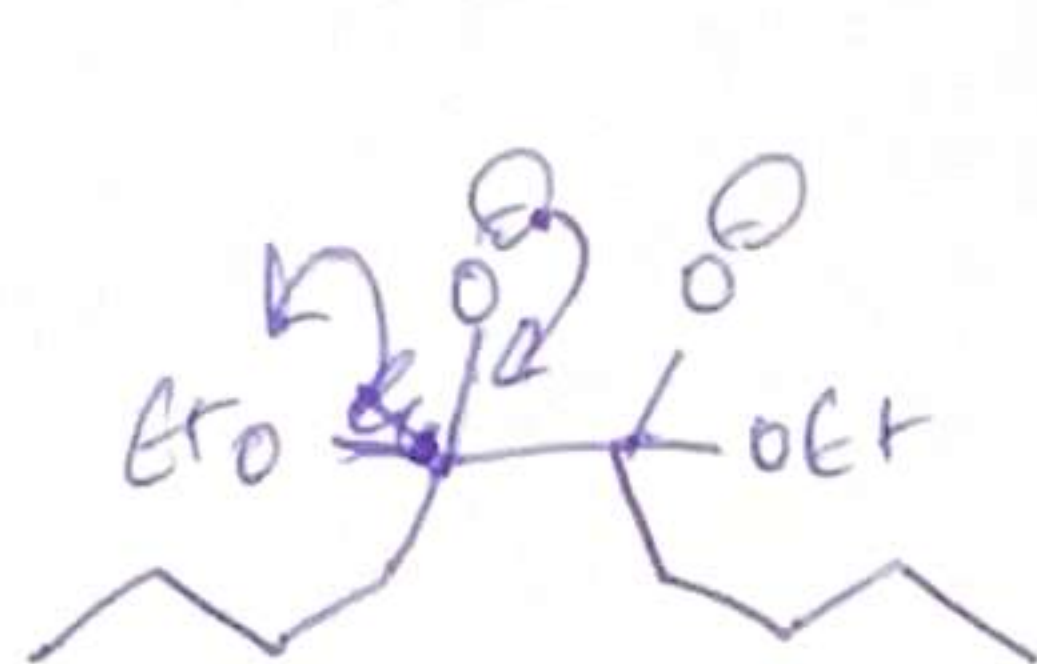
hexyl radical anion



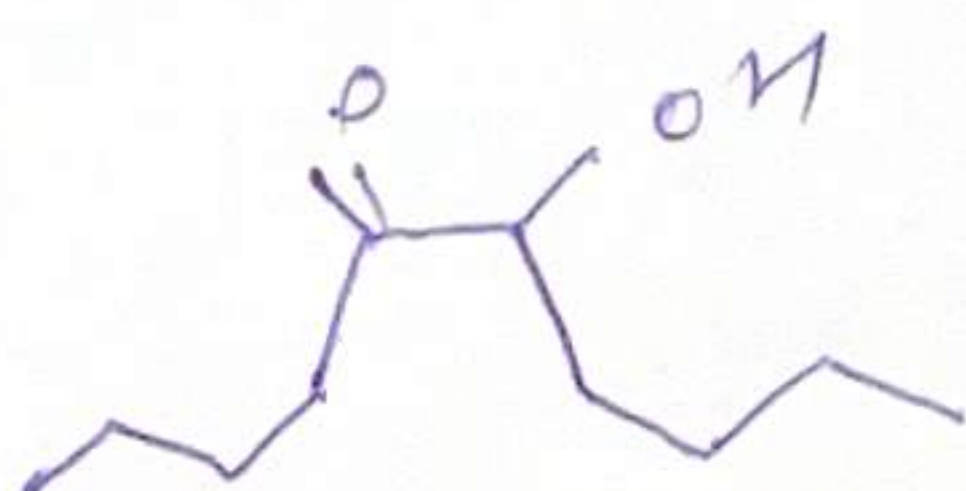
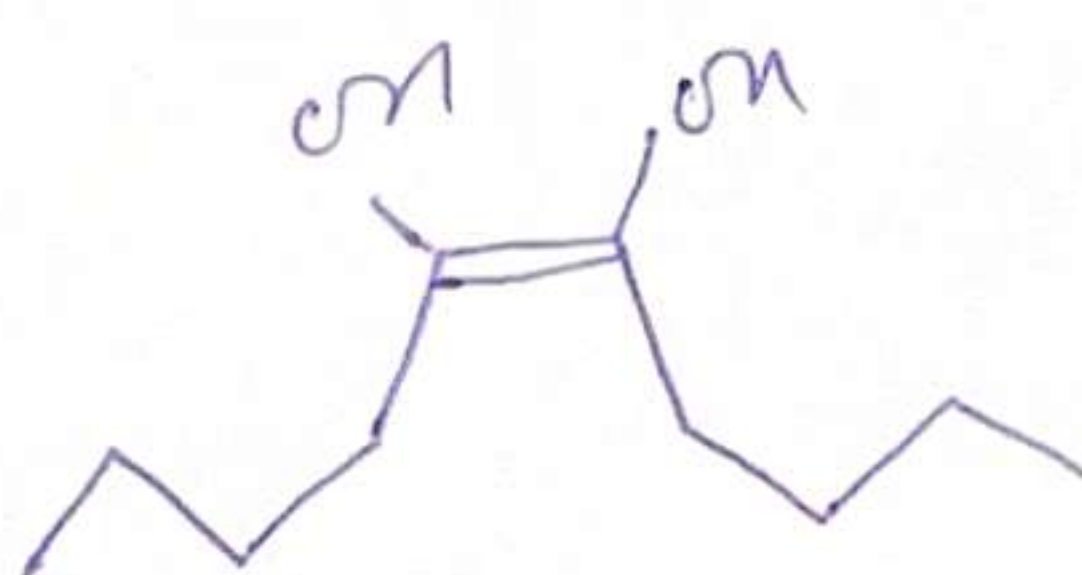
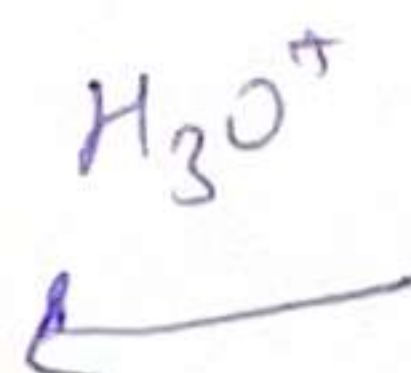
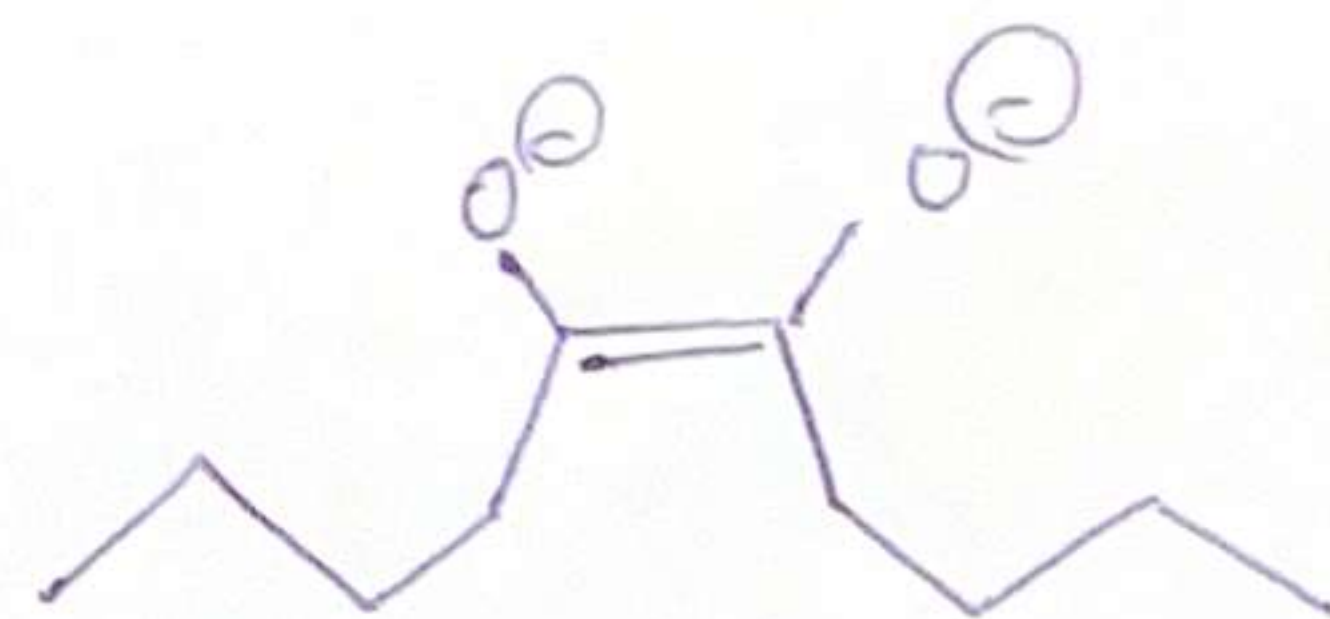
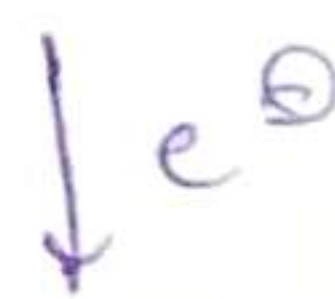
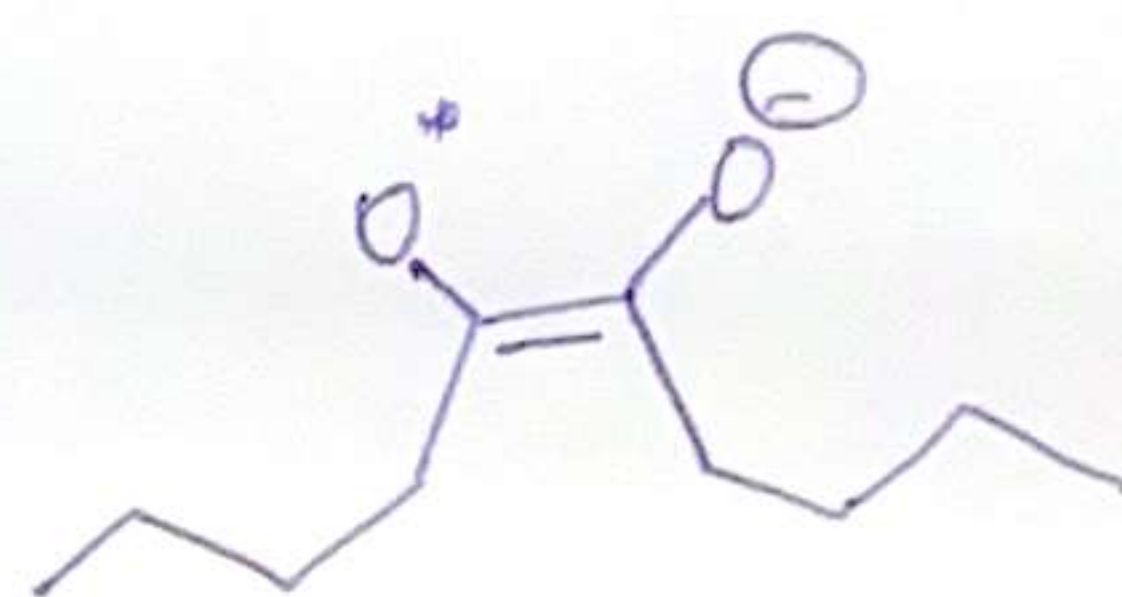
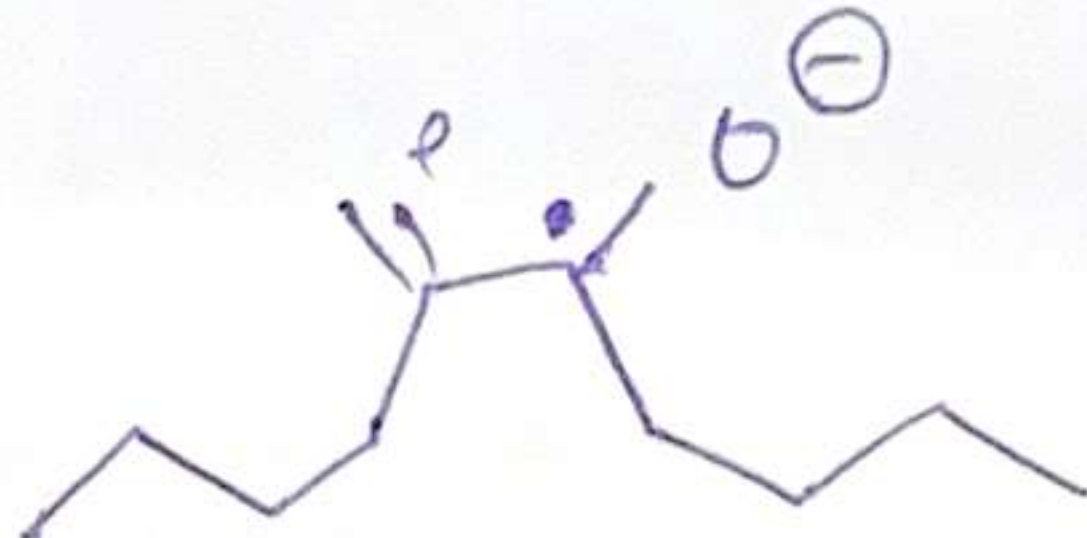
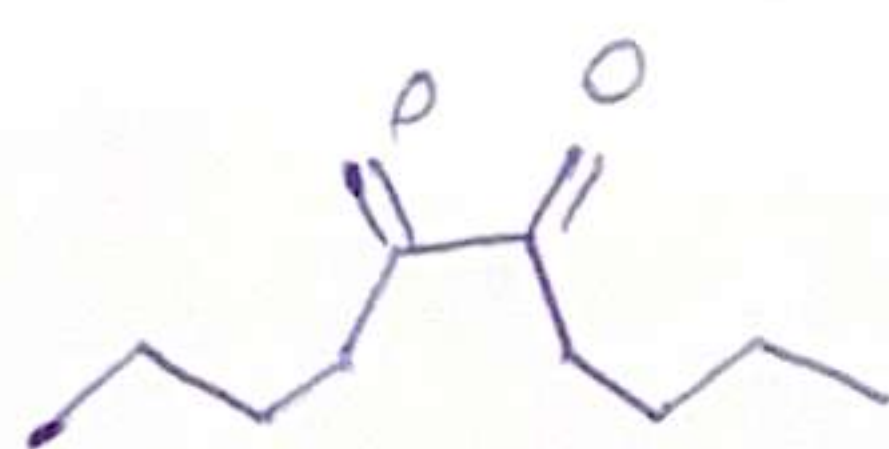
Radical
deprotonation



Unstable
initial Product



1,2-diketone



Acyloin
 α -hydroxy ketone